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What Counts as Good Science? How the Battle for Methodological Legitimacy Affects Public Psychology

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Part of the "boundary work" (Gieryn, 1983) throughout the history of psychology has been to divide the discipline into camps of "basic" and "applied" researchers who take different methodological approaches to construct knowledge. Each "side" has come up with different processes for conceptualizing, constructing, and evaluating the legitimacy of knowledge claims, processes that have implications for applying research insights to practical issues in society. In this article, I review and synthesize research on the history of knowledge construction in both basic and applied psychology, and the implications of their respective methodological practices for their perceived legitimacy. I then discuss how the lessons learned from the past can be leveraged to address the current crisis of confidence in the "credibility revolution" era (Vazire, 2018), as well as the field's perceived legitimacy to external stakeholders. Finally, I end with recommendations for structural changes to improve the credibility and legitimacy of our field's findings as well as their relevance for achieving our public psychology goals.

Public Significance Statement
This article reviews the history of research methodologies in both basic and applied areas of psychology and discusses their implications for the perceived legitimacy of psychological research. After reviewing that history, it proposes policies designed to integrate methods in the field to foster greater credibility and legitimacy in our methods to ensure that our research can be more helpful to the broader public.

Keywords: epistemology, public psychology, ethics, generalizability, metascience

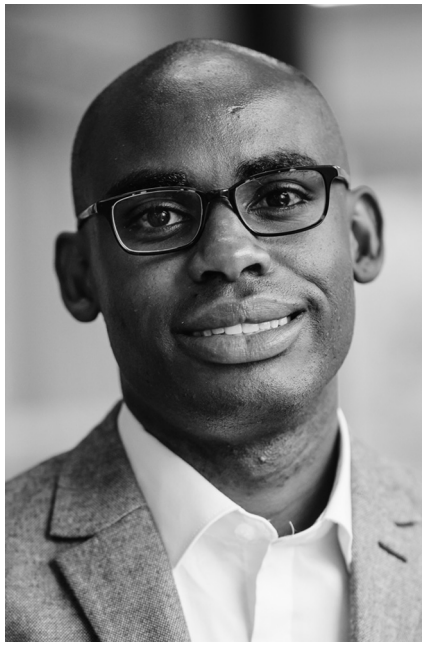
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The Spring of 2020 reignited a debate about an old issue in the discipline of psychology. The emergence the coronavirus disease 2019 (COVID-19) global pandemic and its devastating effects on so many areas of life inspired many psychologists to ask themselves what they could do to bring an end to the pandemic. Some were optimistic that their research could provide some immediate answers, and as a result over 540 new papers about the psychology of COVID-19 were written and shared (via the psychology preprint server PsyArXiv) within the first 6 months of the pandemic (Syed, 2020). Others wanted to help, too, but were less optimistic about the pandemic-readiness of modern psychological research due to how much of that research has been constructed in recent decades (IJzerman et al., 2020). Although the debate about whether and when psychologists should enter the public discourse about the pandemic was new, the underlying issue at the heart of the debate was very old. In fact, research on the history of psychology and science and technology studies suggest that it is an issue that has been with the field since the very beginning (see Hegarty & Rutherford, 2019; Pettit, 2015).

The discourse among psychologists during the pandemic was the latest manifestation of a longstanding battle over *methodological legitimacy:* a battle within the field over which approaches "count" as valid methods of inquiry for generating knowledge about people's thoughts, feelings, and behaviors. During the decade before the pandemic, that battle was largely internal (though it sometimes spilled into the popular press). Psychologists went through a "credibility revolution" (Vazire,

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2018) or "renaissance" (Nelson et al., 2018) that was largely inspired by the replication crisis of the 2010s (e.g., Open Science Collaboration [OSC], 2015) that led many to wrestle with metascientific questions about whether we actually have robust findings about how the mind works or whether what we *thought we knew* rested on shaky methodological grounds (Simmons et al., 2011). But the pandemic made it clear that those metascientific questions were not just abstract intellectual debates among ourselves. Instead, it revealed that they mattered greatly for our ability to responsibly engage in *public psychology*—to use our work to address pressing issues in society (IJzerman et al., 2020; Lewis & Wai, 2021).

In this article I want to discuss that larger issue—the battle for methodological legitimacy in psychology. I want to discuss it not because of the pandemic per se, but because it matters for the larger project of public psychology in both the present and future. It matters for who counts as legitimate psychologists, for what approaches count as legitimate psychological scholarship, for why some things (and people) count whereas others do not. And those things matter because they determine who gets to speak publicly for the discipline and the people we study (or do not study), all of which have important implications for how our science affects people's lives.

To understand those dynamics, we have to engage with the history of the battle, as well as its contemporary manifestations. To be sure, I will not be able to cover the entire history (there is a page limit), but I will attempt to cover enough to clarify why this battle matters now—for who gets to produce knowledge in our field (Roberts et al., 2020), what kind of knowledge gets produced (Cialdini, 2009), how our institutions—both inside and outside of academia—end up operating (Settles et al., 2020), and what that means for the credibility and trustworthiness of our scientific efforts to different publics (Lewis & Wai, 2021).

Who Counts?

The first thing we have to examine is the actors in this battle: who counts as legitimate psychologists, and which research subjects contribute knowledge that is valued in the discipline? Opinions about this topic have been divided over the years between different camps in the field. To explain what I mean, throughout the article I will use heuristics like "basic" and "applied" to delineate the groups. I do this not because I endorse these dichotomies, but because much of the "boundary work" (Gieryn, 1983) in the discipline continues to reinforce them, and that shapes the dynamics I want to discuss: who is seen as legitimate, and who acquires intellectual authority, social power, and influence both within the discipline as well as in the public perception of the discipline (Grzanka & Cole, 2021; Lewis & Wai, 2021; Settles et al., 2020).

On the basic side of the discipline, the people who have "counted" most as being legitimate producers of knowledge have resoundingly been White. Roberts and colleagues (Roberts et al., 2020) recently analyzed over 26,000 papers published between 1974 and 2018 in cognitive, developmental, and social psychology journals and found that most of the editors and authors were White, which affected the makeup of participants since White psychologists study mostly White people, whereas psychologists of color are more likely to study people of color (Roberts et al., 2020). Those findings are consistent with longstanding patterns. Three decades ago, Sears (1986) noted that social psychology studied primarily (White and wealthy) American college students. And 10 years ago, Henrich and colleagues (2010) found that not much had changed in the behavioral sciences more broadly—most samples were drawn from Western, Industrialized, Rich, and Democratic (WEIRD) societies, creating a sampling bias so large that, for example, "in the *Journal of Personality and Social Psychology* . . . a randomly selected American undergraduate [was] more than 4,000 time more likely to be a research participant than is a randomly selected person from outside the west" (p. 63).

This is not always a problem. New ideas have to be tested with *someone*, so the convenient pools of undergraduates, or in more recent decades—pools of Mechanical Turk workers (Anderson et al., 2019)—can be useful places to start. Moreover, since basic research often prioritizes *internal* validity, it can be legitimate to use narrow sampling frames; generalization is not always the point. There are times when psychologists conduct studies to predict real-life behavior, and in those cases elements of external validity—like representativeness in samples and settings—matters, but there are other times when that is not the research goal and external *invalidity* is perfectly legitimate (Mook, 1983).

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The challenge for public psychology is that sometimes research that was designed to be externally invalid is leveraged by psychologists to gain power and influence in real-world scenarios in which external validity *does* matter, and matters greatly; that was the heart of the debates among psychologists during the pandemic (IJzerman et al., 2020). Some assumed that findings generated with basic paradigms would transfer well to more diverse publics affected by the pandemic. If others questioned that *assumption* of universality then, just like in the past, the burden of proof was on them to show that there are in fact differences by factors such as culture, race, and ethnicity (cf., Sue, 1999). This is not a new pattern; it is a continuation of a long trend: basic researchers who study White people have never had to prove universality to claim it; universality and generality are only questioned when the research involves ethnic minority populations (Korchin, 1980; Sue, 1999). This has been the tradition in the discipline, despite even the most ardent defenders of external invalidity long acknowledging that when using research to address real-world problems it is important to attend to representativeness of subjects, settings, and manipulations in our studies (Mook, 1983).

For psychologists on the more applied end of the spectrum—the approach of using narrow sampling frames and making universalistic claims would not be considered legitimate. In fact, those who operate from a more intersectional framework, for instance, would argue that samples must be viewed in terms of their particularity, and findings from those samples interpreted accordingly (Cole, 2009; Grzanka et al., 2017). For example, from an intersectional lens, constructs like gender that basic researchers treat as equivalent across participants in statistical models have different meanings since for example, "the definition of woman is constructed differently for White women and women of color" (Hurtado, 1989, p. 845).

That level of nuance can run up against the value of parsimony held by researchers concerned about issues of statistical power and precision that can—though does not have to—make it difficult to get precise estimates of intersectional effects. This has become an even greater concern in recent years due to calls for more stringent statistical thresholds for declaring findings significant (Benjamin et al., 2018; Smith & Little, 2018). But intersectional researchers would suggest that parsimony is a value in and of itself that we can choose how much weight to assign to it and prioritizing simpler models purely in service of parsimony is a false economy (Cole, 2009; Grzanka & Cole, 2021). Moreover, there are ways to design studies to achieve both goals (see Cokley & Awad, 2013; Smith & Little, 2018; Whitsett & Shoda, 2014).

What these tensions reveal is underlying philosophical differences about whose approaches represent legitimate methods of inquiry. Basic researchers often use methodological frameworks and tools that emerged from European scientific traditions that prioritize detachment, linearity, and universality, and assume that quantification makes the research value neutral, "objective," "good," and legitimate (Gonzalez, 2018; Grzanka & Cole, 2021; Settles et al., 2020). Intersectional psychologists (Cole, 2009; Grzanka et al., 2017), Black psychologists (Cokley & Garba, 2018), Diversity-science psychologists (A. L. Miller et al., 2019), LGBTQ+ psychologists (Hegarty & Rutherford, 2019), and others whose research programs focus primarily on people at the margins of society sometimes use these methods, too (Cokley & Awad, 2013). But they tend to approach the enterprise with a different lens—a lens of greater humility and self-reflexivity in the face of complex historical power dynamics that shaped methodological developments (Grzanka & Cole, 2021). They do not necessarily agree that European approaches are, or should be, the "gold standards" for research to be considered legitimate. In fact, some have argued that differences in cultural values that make it not only inappropriate, but also potentially damaging to use some methods for studying Black psychology (Cokley & Garba, 2018) as well as the psychology of other marginalized populations.

Part of this tension is due to the history of power dynamics in the discipline. Psychology not only sought to craft descriptions of human behavior, but also to develop interventions for improving the human condition (Pettit, 2015). The challenge is that what counted as "improvements" for some, meant marginalization, if not outright extermination for others. Psychology played an active role in the eugenics movement, in torture programs, and our research has consistently been used to reinforce practices of racial superiority and inferiority (A. L. Miller et al., 2019). Not all psychologists are aware of this history; but it is important, as it continues to influence modern scientific discourse. Historical patterns of privilege, oppression, and inequality in the field can influence contemporary research (A. L. Miller et al., 2019), and whether we view researchers from different methodological orientations as legitimate.

What Counts?

The processes I have been discussing matter not only for *who* is considered to be legitimate psychologists, but also *what* counts as meaningful scholarship. On the basic side, the research culture evolved to legitimate research that builds on the previous (homogeneously constructed) literature, and tests theories with samples and methods consistent with dominant approaches (Gonzalez, 2018; Grzanka & Cole, 2021; Settles et al., 2020). That culture delegitimizes approaches that run counter to dominant narratives, as well as approaches that try to incorporate experiential knowledge from groups previously excluded from the historically homogeneous literature (Gonzalez, 2018; Grzanka & Cole, 2021; Stanley, 2007).

To be more concrete, on the basic side, though in the past we built theories with culturally grounded, context-dependent approaches, in recent years—especially since the cognitive revolution—there has been a shift toward

conceptualizing and studying behavior in ways that have stripped away some of the rich and important contextual information (Cialdini, 2009; Kidd, 2002; Pettit, 2015). We have come to *assume* that our measures are objective and universal and sometimes forget that we are asking questions of people with particular histories and experiences that shape their understanding of what we are asking them. Due to these universalistic assumptions, we end up doing things like developing and norming measures on affluent Western populations, assuming those measures capture universally latent factors, and then are surprised when we later learn about the invalidity of those measures elsewhere (see Laajaj et al., 2019, for an illuminating example).

On the applied side, the very idea of universal measures would be viewed with great skepticism, given the long history of scholarship on generality often being limited by cultural and contextual factors. In the world of applied psychology, heterogeneity and context-dependency is the default assumption (Bryan et al., in press; Cole, 2009). Rather than starting with top-down approaches to constructing knowledge, then, applied psychologists often incorporate other methodological approaches that inductively generate knowledge while recognizing and acknowledging that both researchers and participants come to the research setting with subjective experiences that shape what happens in the setting and what we might infer from the research process (Morrow, 2005; Settles et al., 2020).

Although more inductive approaches were valued by basic researchers in the past (see Rozin, 2009), more recently the research projects that have been receiving endorsements for their legitimacy—quite literally in the form of badges on papers (Kidwell et al., 2016)—have been projects adopting hypothetico-deductive models of science (Fidler et al., 2018). The reason for this is journals attempting to reward researchers for adopting open science practices that address the "false-positive psychology" (Simmons et al., 2011) problem that gripped parts of the discipline during the past decade. However, as Grzanka and Cole (2021) note, "attaching visible markers of value to open science practices, rather than other practices (e.g., culturally competent research practices, responsible use of intersectionality), sends an unambiguous signal about what qualities beyond publication itself bring additional value" (p. 16). These kinds of signals are important to attend to: they have downstream implications for what is considered worthy of teaching or learning about in the field (Settles et al., 2020), as well as which scholars are viewed as legitimate and worthy of hiring, retaining, and promoting (Croom, 2017; Griffin et al., 2013).

Why Do Some People and Things Count (and Others Do Not)?

Given the tensions raised about who and what counts in the field, it is worth asking how and why things came to be this way. Although the discipline often presents itself as "objective" and somewhat detached from worldly concerns (Pettit, 2015), the reality is that it has not operated in an intellectual vacuum, nor can it. Psychologists, like the people we study are embedded in broader ecological and social systems that affect our own perceptions, actions, and cognitions (Bronfenbrenner, 1979; Lewis, in press).

The social institutions we are embedded in, and conduct our research in, are highly segregated (Ray, 2019), which matters deeply for our own psychologies (Dupree & Kraus, 2020; Roberts et al., 2020). In the United States where many psychological studies are conducted (Henrich et al., 2010), people are often segregated by race, education, and wealth (Rothstein, 2017). Although Americans like to believe those features are relics of the past (Kraus et al., 2019), that segregation continues in the present day. Those structural differences have meaningful psychological consequences. They affect our social networks (Bayer et al., 2020) and shape our mental representations that guide our judgments and decisions (Lewis, in press).

One consequence of this process is that it has shaped and continues to shape our conceptions of science and scientists. Those in dominant positions within society and within disciplines do not notice the centrality of their positions, and as a result end up assuming that their patterns of thoughts, feelings, and behaviors are "normal" and neutral (Lewis, in press). Thus, it is deviations from those in dominant positions that end up requiring scientific explanation (Bailey et al., 2019; Hegarty, 2017). For instance, White psychologists long assumed their processes to be universal (A. L. Miller et al., 2019) and heterosexual psychologists equated heterosexuality with impartiality and objectivity and deviations from that as "abnormal," and for a long while as psychological disorders (Hegarty, 2013). These assumptions influence the research that gets produced and how it gets disseminated (A. L. Miller et al., 2019). As a result, assumptions of dominant groups who have had the advantages that allow them to be producers of research in the first place get cemented as "ground truths" in the discipline (Dupree & Kraus, 2020; Roberts et al., 2020). Those assumptions shape the discourse until newcomers get opportunities to enter the discipline and question them, as happened when cultural psychologists came along and showed that many principles assumed to be universal were actually highly context-dependent(e.g., Markus & Kitayama, 1991; Markus & Oyserman, 1989).

It is worth noting that this battle for legitimacy is influenced by, and continues to influence, broader politics and struggles for power in the "outside" world (Hegarty & Rutherford, 2019; Pettit, 2015). Psychologists—like other humans—are aware that there is a political strategy of "facting" (Pope, 2012) and aim to construct "good science" that favors, or is at least fits conveniently with, their positions (Hegarty & Rutherford, 2019). For instance, historians of

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science have noted that decisions about how to frame issues or even conceptualize variables in psychology cannot be separated from the moral and political leanings of researchers or the times in which they are conducting the research. For example, appeals to nature became common when discussing LGBTQ issues because "born this way" narratives became the progressive position with respect to sexual behavior; such narratives are frowned upon in research on aggression because genetic explanations still carry racialized connotations that are now taboo among progressives (Longino, 2013; Pettit, 2015). Psychologists have been attuned to how findings might be viewed in policy and politics since at least the 1970s (Pettit, 2015).

Research in the LGBTQ psychology literature has traced this history by examining how the Stonewall riots in New York City have been commemorated, and the implications of that process of commemoration for both science and policy (Hegarty & Rutherford, 2019). Although transgender people of color were central to Stonewall and the movement that emerged from it, issues of race, class, and gender were decentered from the stories of activism that eventually led to the removal of homosexuality as a mental disorder from the Diagnostic and Statistical Manual (Hipp et al., 2019). So, psychologists ended up developing, prioritizing, and presenting knowledge about White cisgender men which was politically effective, but the effect of that process failed to address systemic barriers to equality for LGBTQ+ people of color (Hegarty & Rutherford, 2019; Hipp et al., 2019). Rather than engaging the intersectional issues associated with race and sexuality, psychologists chose to simply use race as an analogy for sexual minority status: there was precedent for objecting to discrimination on the basis of race since race is often conceptualized as an immutable characteristic; that analogy was useful for arguing that discrimination on the basis of sexuality should also be objectionable since sexuality is also something people are "born with" (Hegarty & Rutherford, 2019).

These principles are not unique to LGBTQ+ psychology or other applied areas of the discipline, they also apply to more basic areas of the field. They are relevant to the debates in cognitive psychology about what counts as "intelligence" and the back-and-forth between Black and White psychologists over the measurement of that construct, given its implications for so many critical decisions in life (Cokley & Garba, 2018). They apply to my own area of social psychology and our ongoing debates about which evidence is good enough to tell us whether biases exist in the real world (Cesario, in press) and whether anyone is held accountable for those biases (Daumeyer et al., 2019; Grzanka & Cole, 2021). They apply to research on life span development that conceptualizes when people are formally considered adults—an issue with implications for (disparities in) punishment in the criminal justice system (Wakefield & Wildeman, 2011). The broader point is that these politics—including the act of documenting the perspectives and knowledge of some groups of people (but not others) in the official scientific record—are inseparable from questions of power and identity (Liu & Hilton, 2005).

What Does This Mean for Who Gets to Speak (and How They Engage Publicly)?

The politics just described matter not only for the knowledge that gets created and counted *within* the discipline, they also matter for how that knowledge gets discussed in public (Grzanka & Cole, 2021; Lewis & Wai, 2021). Since the late 1960s there have been recurrent calls for researchers to "give psychology away" (G. A. Miller, 1969). However, the dynamics I have been discussing so far raise questions about what knowledge should be shared, and who should share it, given the broad implications of psychologists engaging with the public.

Psychologists are viewed as the authoritative figures on the topics we study. But that authority comes with important responsibility: we have to remember that it is *our* interpretations of what happened in the studies that get voiced, not our participants—they rarely get to give their input on how we write about them in our papers and speak about them in the press or in meetings with policymakers (Hegarty & Rutherford, 2019; Lewis & Wai, 2021). There is always a danger when we occupy these positions in which we are speaking for others, particularly when we are speaking for those who have less power than we do (Hegarty & Rutherford, 2019).

On the basic side of the discipline, this dynamic often remains invisible: the assumption is that the research is done in an objective and detached way and it does not matter who speaks (Hegarty & Rutherford, 2019). The data "speaks for itself" and the faces behind it are considered irrelevant. Of course, the people who often get to speak are those who have optimized their performance on the metrics used to determine whether they are "famous yet" (Sternberg, 2016) and worthy of employment, tenure, and promotion. The fact that there are gender (Eagly & Miller, 2016) and racial (Syed, 2017) biases in opportunities to gain such eminence and fame is conveniently overlooked, as are the limitations of those metrics.

On the applied side, part of the culture of communication is to acknowledge reflexively that data never speaks for itself; we choose what to emphasize and de-emphasize to showcase the things that *we* find interesting, and that those interests are guided in part by our own experiences (Grzanka & Cole, 2021; Jones-Rooy, 2019; Lewis & Wai, 2021). There is greater emphasis placed on continually reflecting on how one's own life experiences influence the research and interpretation, and encouragement to be open and transparent about that as we share findings (A. L. Miller et al., 2019; Pettit, 2015). This reflection process is

considered essential: acknowledging how researchers' social positions influence their interpretations is part of the process of conducting legitimate applied research (Morrow, 2005).

The Convergence of Methodologies and Their Implications for Collective Legitimacy

So far, I have discussed the differences in approaches taken by basic versus applied researchers in their respective quests to conduct and disseminate what they each perceive to be legitimate science. I have also discussed why, given the history of who comprised those respective parts of the discipline and the methods they used, each side may not consider the others' work to be as legitimate as their own, and worthy of public dissemination and discussion. These divisions are not new; they have been in the discipline for generations and have even been formalized in some ways. During the review process for this article one piece of history I was reminded of was the distinctions between the American Psychological Association community (more applied) and Association for Psychological Science community (more basic) and what those distinctions mean for even the kinds of research articles their respective journals are interested in publishing. It was a reminder of the epistemic priorities of different parts of the discipline, and how those priorities go on to shape who is represented (or absent) from different parts of the discipline as a result (Ledgerwood et al., 2021; Settles et al., 2020).

These divisions are not inevitable, however. One thing that struck me as I reviewed articles from across the spectrum of the discipline to write this paper was that the current moment in the field's history may be an opportunity for disparate parts of the discipline to learn more from each other and work together to co-construct knowledge in ways that our disciplinary collective might consider to be legitimately good science. Why have I become so optimistic?

The replication crisis of the 2010s created a crisis of confidence about the legitimacy of research that was being conducted, mostly in the basic side of psychology (OSC, 2015; Simmons et al., 2011; Vazire, 2018). Metascientific investigations found, for instance, that we were only able to replicate somewhere between one-third and two-thirds of findings (e.g., OSC, 2015). Moreover, we also learned that many published methods and measures were not properly validated (Flake et al., 2017), making it difficult to know what they actually mean. Overall, one takeaway from that era is that many basic models were not constructed in ways that allowed us to form good representations of psychological processes (Navarro, 2019), making it difficult to come up with good theories (Devezer et al., 2020). That creates a public psychology problem, because if the evidence behind the theories are shaky, it makes it difficult to translate those theories into practical applications, especially during moments of crisis—moments like a global pandemic (IJzerman et al., 2020).

The discipline has been trying to tackle some of these problems by proposing new solutions like making statistical thresholds more stringent for declaring a finding "significant" (Benjamin et al., 2018) and pushing for open science practices such as preregistration, open materials, and open data to foster greater credibility (Nelson et al., 2018). But these solutions on their own do not solve many of the fundamental issues, and metascientific studies have provided evidence for that. For example, in a recent study entitled "Many Analysts: One Data Set" researchers sent 29 research teams the same dataset to answer the same question: are soccer referees more likely to give red cards to dark-skin-toned players than light-skin-toned players (Silberzahn et al., 2018). Because the research question and dataset are identical, researchers subscribing to more universalistic ways of thinking might expect that all teams would converge on the same process and answer. That is not what the researchers found. The 29 teams analyzed the data 21 different ways; 20 teams found a statistically significant and positive effect, nine found no significant relationship, and effect sizes ranged from .89 to 2.93 in odds-ratio units (Silberzahn et al., 2018). What those findings suggest is that the answer to the research question depends, in part, on which team happened to analyze the data, and their experiences that led them to make the assumptions that guided their analytic decisions (Lewis, 2020).

Researchers from more context-dependent traditions are less surprised by these kinds of findings. It is taken as a given that histories and life experiences shape the ways in which research is conducted (A. L. Miller et al., 2019). That is why statements of reflexivity (Morrow, 2005) and triangulation across methods are more normative in those research traditions (Settles et al., 2020). This underscores the importance of diversifying the teams of people doing research and the approaches brought to bear in answering societally important research questions—greater diversity brings novel and important perspectives for understanding psychological phenomenon (Dupree & Kraus, 2020; Ledgerwood et al., 2021; Roberts et al., 2020). More pluralistic approaches and ways of thinking may alleviate some of the methodological angst that has been plaguing the discipline over the past decade, and perhaps lead us to gain more accurate understandings of the people and topics we are studying. It can also allow us to do more actionable research in partnership with diverse publics.

This is something that has become clearer to me in my own work on public psychology projects over the past few years. I have spent much of the past year on multiple "behavioral science" task forces working with different levels of government on the COVID-19 pandemic response (e.g., Barsade et al., 2021). My role on each of those task forces is usually to help figure out which research is relevant and actionable to improve

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equity in pandemic outcomes. Doing that well requires a mix of both the basic and applied approaches. On the topic of building confidence in vaccines, for example, one theme that often comes up from the basic science of persuasion is to use normative influence to persuade people to get a vaccine. This approach can be helpful, but previous applied research on normative interventions in health has documented that histories of marginalization matter for how those interventions need to be operationalized in practice. What resonates and works well for high powered majority group members can be quite different than what works for low power minority group members (Durantini et al., 2006). Without inputs from both sides of the discipline, it is difficult to develop relevant and actionable insights to improve outcomes, especially equity in those outcomes.

The pandemic is one social crisis that has brought this need to collaborate across methodological approaches to the fore, but it is not the only one. These collaborations also matter for the looming environmental crisis that lies ahead, and the role that psychologists and other social scientists might play in that crisis. Over the past few years, my colleagues and I have been examining perceptions, actions, and cognitions about environmental issues in the United States, and heterogeneity in those psychological processes among subgroups of Americans (Lewis et al., 2020; Pearson et al., 2018; Song et al., 2020). To provide a bit of background, the environmental movement and environmental and climate science community in the United States is *very* White; it is one of the least diverse scientific disciplines in the United States (Pearson & Schuldt, 2014).

That lack of diversity has shaped the questions that have been asked, methods of inquiry, and conclusions drawn from studies—conclusions such as the perception that racial and ethnic minority and low-income people are not as concerned about the environment as wealthy White people (Pearson et al., 2018). We wondered whether those conclusions would hold if we broadened the pool of participants in our studies, as well as our methods of inquiry for examining environmental psychology processes. What we have learned is that, in part due to the structural forces I outlined earlier, what you find depends on *who* you study, and *how* you study them. Specifically, across several studies that have used a mix of qualitative and quantitative methodologies as well a variety of sampling strategies to diversify the samples of people we are study, we have consistently found that race, ethnicity, and socioeconomic status shape Americans' environmental psychology (Song et al., 2020). The differences in life experiences that various groups of Americans have had with respect to environmental issues have shaped their stereotypes and metaperceptions about who is concerned about environmental issues (Pearson et al., 2018) as well as the content of what even "counts" as environmental issues in the first place (Lewis et al., 2020; Song et al., 2020). So, the psychology of environmentalism manifests very differently among different subgroups of people *even within the same country*, meaning our interpretations of constructs and measures, and communication about environmental psychology must be calibrated accordingly.

Why does this matter? One of the primary goals of research psychologists is to construct reliable and valid knowledge that teaches people about how the mind works, and how the mind influences our behavior and vice versa. What these findings remind us though, is that without diversity in our teams and samples, and without expanding our methodological approaches to include a variety of methods that allow participants to tell us more about what they think, it becomes difficult to know whether and under what conditions our constructs are defined and operationalized properly, and whether those findings should be applied to decisions that affect people's lives (IJzerman et al., 2020; Ledgerwood et al., 2021; Lewis et al., 2020).

These insights from the applied side of psychology are consistent with new recommendations from basic research methodological reformers to be more explicit about our assumptions and justifications of methodological choices, and recommendations to be more transparent about the constraints on the generality of empirical research findings (Simons et al., 2017). In other words, the "new" methods that basic researchers have come up with as approaches and standards for increasing the credibility and legitimacy of their research are functionally equivalent to the approaches that applied researchers have been developing and refining for decades. The reflexivity and transparency that have always been required for trust in applied research, turns out to also be required for trust and legitimacy in basic research, too.

Structural and Cultural Shifts to Move Us Forward

Now that it is becoming clearer that the field's collective legitimacy and relevance for the public can be enhanced by combining and integrating the methods each side has been developing, we have to consider how the sociology of the field and the incentives within it might need to change to move the field forward (Ledgerwood et al., 2021).

With respect to representation in the field, there has long been evidence that representation matters for producing applicable knowledge (Cornesse et al., 2020; Mook, 1983), there has just been little incentive to actually diversify samples—it is not required or even valued in the mainstream journals (Korchin, 1980; Sue, 1999). Moreover, the people who have paid most attention to this issue—minoritized scientists—have not been in the positions (e.g., editors of journals) that enable them to direct more field-wide attention to these issues (Roberts et al., 2020). I am hopeful that as we grapple with how critical representation is for the construction of actionable knowledge, there will be new emphasis placed on this in our journals, and in our hiring practices. I am aware, however, that hope alone is

not enough. Research on diversifying organizations has documented the importance of changing organizational structures to achieve these changes (Onyeador et al., 2021). Specifically, implementing accountability structures has been shown to be one of the most effective ways to achieve greater equity in organizations (Kalev et al., 2006). In academia that might mean not only diversifying the editorial boards—in both basic and applied journals—but also auditing the journals regularly and adjusting decision-rules iteratively to move the field forward.

In addition to shifting structures to bring more marginalized voices into the field to engage in public psychology projects, we will also have to ensure they are *truly* supported and valued, otherwise they are unlikely to stay (Settles et al., 2020). As Grzanka and Cole (2021) note, psychologists sometimes celebrate those who do high-profile, public-facing work, but gate-keeping at virtually all levels continues to sustain the status quo and discourage deviation from the mainstream work of the field.

There are a few potential avenues for intervention. One concrete area to focus on is the funding landscape, given its implications for both who produces science and what science gets produced. Scientists often need grants to cover the costs of their work, and institutions consider funding in hiring decisions. That is why it is troubling that several studies have found systematic racial biases in grant funding (e.g., Ginther et al., 2011). For instance, one recent study that analyzed 157,000 proposals submitted to the US National Institutes of Health (NIH) found that Black scientists are less likely to receive NIH funding due to the topics they are interested in studying (Hoppe et al., 2019). Black applicants are more likely to propose research approaches using community interventions and topics such as health disparities, and those topics are less likely to be funded than topics proposed by White applicants; the researchers estimate that topic choice accounts for 21% of the funding disparities (Hoppe et al., 2019).

Because Black scholars and other psychologists of color are more likely to conduct research with communities of color (Roberts et al., 2020), the NIH and other grant-makers' decisions to systematically defund the scholarship of scientists of color not only undermines the demographic representation of the scientific workforce, it also undermines representation of more diverse cross-sections of the population in the body of scientific evidence. That limits the ability of the science that is produced from being useful for application in the world. Funders can, and should, change their operating procedures to rectify these patterns; Stevens and colleagues (2021) outlined a set of recommendations for doing so.

These changes are essential not only for the advancement of science, but also for the lives our science affects. For example, in their synthesis on how impacts of policies are communicated, Brick and colleagues (2018) note that policy impacts often vary by geography, culture, and demographics. Because of this, who is in the studies that are used to justify policy decisions matters greatly. For example, in Europe there have been proposals to ban the sale of new gasoline and diesel cars due to the local environmental benefits (DEFRA, 2017). These policies will surely improve the health and well-being of Europeans. However, one consequence of the European gasoline car bans is that they increase the demand for lithium-ion batteries containing cobalt, half of which comes from mines in the Democratic Republic of the Congo (Yager, 2017). Nongovernmental organizations like Amnesty International (2016) have reported widespread health risks to tens of thousands of Congolese miners—including many children—and the United Nations has reported major environmental damage (UNEP, 2011) due to this. Thus, if policymakers act based on research constructed with the public opinion of Europeans who are only aware of the benefits to their lives, these policies will seem like good things; if they consider the Congolese people whose lives they are sacrificing to save European lives, they might reach different conclusions (Brick et al., 2018).

This highlights the importance of shifting our knowledge construction practices toward methods that include the representation of the range of people whose lives will be affected by our research. If the research is only constructed with dominant group perspectives that might be fine for the purposes of internal validity (Mook, 1983), but it can become dangerous if that same research is used for public psychology projects. This is particularly problematic given the ways in which research is used to justify "change-making" in the modern world. Girdharadas (2019) illustrated this problem in his book *Winners Take All: The Elite Charade of Changing the World.* In the book, he interviewed a series of wealthy and powerful people who wish to change the world, as well as high-profile psychologists whose research and speeches they turn to for guidance on how to achieve the global changes they desire. One of the things I found most striking from the interviews was the way researchers communicated about findings when talking to powerful change agents. When given the opportunities to speak on large stages (like the TED conference), psychologists often shared quick-fix solutions found in their studies with nonrepresentative samples, leading those change-makers to believe that those are effective solutions for social change, rather than addressing broader systemic issues in society.

Of course, not all psychologists engage in public scholarship in those ways, but it is nevertheless important to note this because in matters of science communication the actions of the few matter for the many (Lewis & Wai, 2021). Thus, if we wish to engage in responsible public psychology, it is important to build structures and cultures that facilitate the production of knowledge that seriously grapples with the epistemic questions of how our knowledge is constructed—who is included and excluded, what methodological practices were used and what those decisions mean for the legitimacy of our findings, and the appropriateness of broadly

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disseminating them to the world (Grzanka & Cole, 2021; Navarro, 2019; Settles et al., 2020).

This final point is one that began weighing more heavily on me a few years ago when my colleagues and I began working on a series of community-engaged environmental psychology projects (Lewis et al., 2020). To our surprise, many of our participants thanked us for coming to their communities and allowing them to participate in our studies; that is not something that typically happens in our studies of undergraduates. Our community participants appreciated the opportunity to participate because, in their words: "no one usually cares what [we] think."

If this is a common sentiment, then it reflects a problem for public psychology and other disciplines that wish for their work to have larger public impact. It suggests that the scientific community has taught marginalized communities the same lesson that I learned from an op-ed writing workshop a few years ago: that "some people narrate the world; other people have their world narrated for them" (The Op-Ed Project, 2017). When we exclude marginalized and minoritized communities because we do not view them as having anything (novel) to contribute to what we consider to be legitimate knowledge, or to our institutions (Settles et al., 2020), we are telling them that there are some people who get to be part of the scientific record, and others who do not—and they are always in the latter category.

But this perpetual exclusion from the research process has important downstream consequences for what society does with that research. As Squire (2015) noted: "historical exclusivity often has a way of turning into present and institutionalized tragedy. Whose story gets told matter." Who gets to produce our studies, and how they get to produce it, matters. Who gets to be part of our studies matters. When we make decisions about who to include, we are implicitly making decision about whose minds matter to us—whose perspectives we consider to be legitimate and worthy of a place in our field. And we have to remember that, because those decisions have important implications for the knowledge we create, and the policies and practices that are developed as a result of that knowledge.

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